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Preface

The field of microwave engineering has undergone a radical transformation in recent years as the Defense and Government work effort that once dominated the industry is now exceeded by commercial wireless efforts. The explosive growth of commercial wireless markets has not only altered the Defense/non-Defense work balance, but also brought about significant shifts in the perspective and focus of working microwave and RF engineers. Engineering emphasis has changed from optimum performance design to design for manufacturing, from one-of-a-kind parts to high volume production, from performance at all cost to minimum cost with acceptable performance, from widest possible bandwidth to regulated narrow band, etc. Even engineers and researchers working in the traditional high performance, low volume microwave markets have seen a shift in emphasis of their work as every effort is made to reduce cost through the re-use of commercial-off-the-shelf (COTS) parts. Although the physics and mathematics of microwave and RF engineering is the same, the job of the typical microwave engineer has changed dramatically. The modern microwave and RF engineer is expected to be knowledgeable of customer expectations, market trends, manufacturing technologies, and factory models to a degree that is unprecedented in the history of RF/microwave engineering. Unfortunately, the 40+ years of close association of the microwave industry solely with Defense/Government agencies has left a legacy of microwave literature that is narrowly focused on high performance, low volume applications and is deficient in many areas of critical importance today. This handbook comes at a time when there is an emerging interest in a broader range of microwave material with more balance than that which has been previously available.

The purpose of *The RF and Microwave Handbook* is to provide a single-volume comprehensive reference for modern microwave and RF engineers. The articles that comprise the handbook provide important information for practicing engineers in industry, government, and academia. The intended audience also includes microwave and other electrical engineers requiring information outside of their area of expertise as well as managers, marketers, and technical support workers who need better understanding of the fields driving and affected by their decisions.

The book is organized into nine chapters, with all but the first chapter consisting of several articles on related topics. The first chapter consists of a single introductory article that provides important definitions and spectral information. Two appendices containing useful information for practicing microwave engineers are also included. By design, there is some overlap of information presented in some of the chapters. This allows the reader to investigate a topic without needing to make numerous references to other portions of the handbook. A complete table of contents is presented at the front of the book to provide the reader with a means of locating desired information as easily and rapidly as possible. Finally, all of the articles provide the reader with additional references to related expert literature.

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Acknowledgments

Developing a handbook like this one is a big job — much bigger than I originally anticipated. This handbook would simply never have been completed if it were not for the efforts of the managing editor, Janet Golio. Her focus, persistence, software expertise, and organizational skills were essential to the project. Her efforts can be measured in the nearly 10,000 pieces of email correspondence she handled, the 100+ telephone conversations she initiated, or the tracking of approximately 80 articles from initial contact to commitment to receipt to review through final modification and submission. Yet all of these metrics combined do not completely capture the full magnitude of her contribution to this handbook. I cannot offer enough gratitude to compensate for the many long evenings she spent on this project.

I am also significantly indebted to the Handbook Editorial Board. This Board contributed to every phase of the handbook development. Their efforts are reflected in the organization and outline of the material, selection and recruitment of authors, article contributions, and review of the articles. Their labors were essential to the project and I am happy to acknowledge their help.

And, of course, I must thank the handbook professionals at CRC Press. Richard Dorf, Editor for the CRC Electrical Engineering Handbook series, and Ron Powers, President of Book Publishing, were instrumental in getting this project started. Special thanks is extended to Nora Konopka, Acquisitions Editor, who has worked most closely with the project during article development and has been more patient and encouraging than I deserve. Finally, Helena Redshaw, Production Manager, has taken the stacks of manuscripts, disks and CDs, identified and added the missing bits and pieces, and turned them into a book. Thanks also to all the CRC staff that I have not had the pleasure to work closely with, but who have contributed to this effort.

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Editor-in-Chief

Mike Golio received his B.S.E.E. degree from the University of Illinois in 1976. He worked designing microwave oscillators and amplifiers for electronic counter-measure applications for two years before returning to school to complete his M.S.E.E. and Ph.D. degrees at North Carolina State University in 1980 and 1983, respectively. After graduate school, he worked as an Assistant Professor of Electrical Engineering before returning to industry. He spent five years developing MMIC circuits, tools, and techniques for microwave satellite applications prior to participating in the establishment of a GaAs fabrication facility. The GaAs facility was constructed to manufacture devices and circuits to serve commercial wireless markets — including cellular telephone, pager, and wireless LAN businesses. His role shifted from individual technical contributor to Group Manager during this time. He is currently the Director of the Rockwell Collins RF/Power Design Center tasked with identifying, acquiring, developing, and implementing emerging RF and power conversion technologies. The Center provides design expertise in the radio frequency (RF), microwave, and millimeter wave domain. This includes receivers, transmitters, antennas, frequency control circuitry, and power supplies for commercial and military avionics applications.

Dr. Golio's research has resulted in 15 patents and over 200 professional presentations and publications. His technical publications include four book chapters, one book, and one software package. He was selected to serve as the IEEE Microwave Theory and Techniques-Society (MTT-S) Distinguished Microwave Lecturer from 1998 to 2000. He is currently co-editor of the I*EEE Microwave Magazine,* to which he contributes a regular column. He serves as a member of the MTT-S Administrative Committee, has organized several professional workshops and panel sessions, and served as a member of the Technical Program Committee for several professional conferences. He was elected Fellow of the IEEE in 1996.

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Managing Editor

Janet R. Golio received her B. S. in Electrical Engineering Summa Cum Laude from North Carolina State University in 1984. Prior to that she was employed by IBM. Between 1984 and 1996 she worked at Motorola in Arizona. Since 1996, she has been at Rockwell Collins in Cedar Rapids, IA. She holds one patent and has written four technical papers. When not working, she actively pursues her interests in archaeology, trick roping, and country western dancing. She is also the author of young adult books, *A Present From the Past* and *A Puzzle From the Past.*

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Advisory Board

Peter A. Blakey obtained a B.A. in Physics from the University of Oxford in 1972, a Ph.D. in Electronic Engineering from University College London in 1976, and an MBA from the University of Michigan in 1989. Between 1972 and 1983 he worked on the theory and simulation of microwave semiconductor devices and circuits, with an emphasis on the impact of non-local transport phenomena. Between 1983 and 1995 he worked on the development, application, and commercialization of technology CAD software for silicon VLSI applications. Since 1995, he has been the manager of Predictive Engineering in the GaAs Technology Department at Motorola.

Lawrence P. Dunleavy received the B.S.E.E. degree from Michigan Technological University in 1982, and the M.S.E.E. and Ph.D. degrees in 1984 and 1988, respectively, from the University of Michigan. He has worked in industry for E-Systems and Hughes Aircraft Company and was a Howard Hughes Doctoral Fellow. In 1990 he joined the Electrical Engineering Department at the University of South Florida, where he now holds the title of Associate Professor. From August 1997 to to August 1998 he enjoyed a 1-year sabbatical research appointment with the Microwave Metrology Group of the National Institute of Standards and Technology (NIST) in Boulder, CO. His current research interests are in the area of accurate microwave and millimeter-wave measurements, measurement-based active and passive component modeling, MMIC design, and wireless systems characterization and CAD. Dr. Dunleavy is a Senior Member of IEEE, is very active in the IEEE MTT Society, and has been an ARFTG member since 1986.

Dr. Dunleavy has served as local host and technical program chair for ARFTG, as MTT/ARFTG liaison, and as conference chair for various conferences. He has also served ARFTG as the Short Course Coordinator from 1995 to 1998, as Director from 1998 to present, as Chair of the Education Committee from 1997 to present, and as a member of the Executive Committee from 1998 to present.

Jack East received his B.S.E., M.S., and Ph.D. degrees from the University of Michigan. He is presently with the Solid State Electronics Laboratory at the University of Michigan conducting research in the areas of high speed microwave device design, fabrication, and experimental characterization of solid state microwave devices, nonlinear device and circuit modeling for communications circuits and low energy electronics, and THz technology.

Patrick J. Fay is an Assistant Professor in the Department of Electrical Engineering at the University of Notre Dame. Dr. Fay received the B.S. degree in electrical engineering from the University of Notre Dame in 1991, and the M.S. and Ph.D. degrees in electrical engineering from the University of Illinois at Urbana-Champaign in 1993 and 1996, respectively. From 1992 to 1994, he was a National Science Foundation Graduate Research Fellow. After graduation, Dr. Fay served as a visiting assistant professor in the Department of Electrical and Computer Engineering at the University of Illinois at Urbana-Champaign in 1996 and 1997.

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Dr. Fay joined the University of Notre Dame as an assistant professor in 1997. His educational activities have included the development of an advanced undergraduate laboratory course in microwave circuit design and characterization, and graduate courses in optoelectronic devices and electronic device characterization. His research interests include the design, fabrication, and characterization of microwave and high-speed optoelectronic devices and circuits for ultra-fast analog and digital signal processing, wireless communications, and fiber optic telecommunications. His research also includes the development and use of micromachining techniques for the fabrication of active and passive microwave components.

Dr. Fay is a member of the IEEE Microwave Theory and Techniques Society, the Solid-State Circuits Society, and the Electron Device Society. He was awarded the Department of Electrical Engineering's IEEE Outstanding Teacher Award in 1998–1999, and has served as a consultant for the Division of Electrical and Communications Systems at the National Science Foundation.

David Halchin received the Ph.D. degree in electrical engineering from North Carolina State University in 1987. From 1987 to 1989 he was an Assistant Professor in the ECE Department of Santa Clara University in Santa Clara, CA. His main interests in teaching and research were device modeling and microwave design. From 1989 until 1996 he was with Motorola Corporate Research and SPS in Phoenix, AZ. While at Motorola, he worked on a wide variety of projects. These included microwave and millimeter wave characterization of MESFETs and PHEMTs, modeling of PHEMTs and MESFETs, load pull measurements, device (MESFETs and PHEMTs) engineering and process development. From 1996 until 1997 he worked for Rockwell International in Newbury Park, CA as the head of a team tasked to develop and implement into production a linear MESFET and PHEMT for wireless PA applications. He rejoined Motorola SPS in 1997 for 1 year and again worked with the PHEMT/MESFET device development team in Phoenix. In 1998 he joined RF Micro Devices in Greensboro, NC working as a PA designer and a group leader in the area of WCDMA applications. He is a member of the Technical Program Committee for the GaAs IC Symposium, a reviewer for Transactions on Electron Devices and a member of the Editorial Board for Microwave Theory and Techniques and an IEEE member. He has 3 issued U.S. patents and several pending.

Roger B. Marks received his B.A. in Physics in 1980 from Princeton University and his Ph.D. in Applied Physics in 1988 from Yale University. Following a postdoctoral appointment at the Delft University of Technology (The Netherlands), he began a professional career with U.S. Commerce Department's National Institute of Standards and Technology (NIST) in Boulder, CO.

In 1998, Dr. Marks initiated an effort to standardize fixed broadband wireless access within the IEEE 802 LAN/MAN Standards Committee. This effort led to creation, in March 2000, of the IEEE 802.16 Working Group on Broadband Wireless Access. With Dr. Marks as Chair, 802.16 has grown to include three projects and hundreds of participants from over 100 companies.

A Fellow of the IEEE and the author of over 70 journal and conference publications, Dr. Marks has received many awards, including an IEEE Technical Field Award (the 1995 Morris E. Leeds Award). He has served in numerous professional capacities. In the IEEE Microwave Theory and Techniques Society, he served on the Adminstrative Committee and as Distinguished Microwave Lecturer. He developed the IEEE Radio and Wireless Conference (RAWCON) and chaired it from 1996 through 1999. He served as Vice Chair of the 1997 IEEE MTT-S International Microwave Symposium.

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Dr. Marks lives in Denver, CO with his wife, Robbie Bravman Marks, and their children, Dan and Amy. His drumming can be heard on the RAWCON theme song and in a recording by the band Los Lantzmun.

Alfy Riddle operates Macallan Consulting, which specializes in solving RF and microwave circuit design problems as well as selling Mathematica-based design software. Dr. Riddle also teaches a course in electronic noise at Santa Clara University. He received his Ph.D. from North Carolina State University in 1986 for work on oscillator noise. After spending several years employed in the electronics industry, he began Macallan Consulting in 1989. When he is not working, he can be found on the tennis courts, hiking in the Sierras, taking pictures with an old Leica M3, or making and playing Irish whistles.

Robert J. Trew serves as Director of Research for Department of Defense (DoD). He has management oversight responsibility for the $1.2 billion yearly basic research programs of the Military Services and Defense Agencies. Dr. Trew has extensive experience in industry, academia, and government. He received the Ph.D. degree in electrical engineering from the University of Michigan in 1975. His industrial experience includes employment at General Motors Corp. in Michigan and Watkins-Johnson Company in California. In addition to his present DoD assignment, he worked for the U.S. Army Research Office as a program manager for five years. Dr. Trew was on the faculty of North Carolina State University where he served as Professor of Electrical and Computer Engineering from 1976 to 1993. From 1993 to 1997 he was George S. Dively Professor of Engineering and Chair of the Department of Electrical Engineering and Applied Physics at Case Western Reserve University. He was a Visiting Professor at the University of Duisburg in Germany in 1985. Dr. Trew is a Fellow of the IEEE and serves on the Microwave Theory and Techniques Society ADCOM. He is currently serving as Chair of the Publications Committee and was the Editor-in-Chief of the *IEEE Transactions on Microwave Theory and Techniques* from 1995 to 1997. He is currently co-editor of the new *IEEE Microwave Magazine*. Dr. Trew was the recipient of the 1998 IEEE MTT Society Distinguished Educator Award. He received the 1992 Alcoa Foundation Distinguished Engineering Research Award and a 1991 Distinguished Scholarly Achievement Award from NCSU. He was a recipient of an IEEE Third Millennium Medal award, and is currently an IEEE Microwave Distinguished Lecturer. Dr. Trew has 3 patents and has published more than 140 technical articles, 13 book chapters, and given over 250 scientific and technical presentations.

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Appendix A Mathematics, Symbols, and Physical Constants

Appendix B Microwave Engineering Appendix *John P. Wendler*

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